



## Solar Energy And EV Charging Infrastructure

e-Mobility and electric vehicle (EV) charging infrastructure are some of the key focus areas, and relentless research is being done to address high performance in all functional blocks



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**W**ith the push to move to electric mobility at a national scale, and with the government being very enthusiastic about deployment of electric vehicles (EVs), the electronics industry can expect to see a lot of activity around indigenous development and manufacturing. While EVs are being worked upon by major OEMs, an ecosystem for development of chargers, charging stations, and software and cloud services is steadily being built. Established companies as well as various startups have started working on these areas, and results are starting to show. However, there is still a lot of opportunity to make the electronics side of it even better.

The government, with the help of BIS, ARAI, EESL and other bodies, has released technical specifications on charging stations. Some of the original specifications like AC-001 and DC-001 have been developed and charging stations have been deployed at select locations. Newer guidelines require charging stations to be

equipped with multi-standard chargers, namely, AC type 2, CCS and CHADEMO, in addition to lower power AC and DC-001 ones. However, these systems are reliant entirely on the grid, and are subject to real estate availability at prime urban and semi-urban locations. Besides, the question of the grid being ready and equipped for these added loads still remains.

This is where solar energy and storage comes into the picture, to not only supplement the grid but to also work standalone at feasible locations across the country. Fortunately, India has seen successful solar deployments and enjoys abundance of solar energy due to its geographic location. One-time installation and capital expenses work well for at least 20 to 25 years, with the ROI taken care of in a few years. Energy input thereafter becomes virtually free.

A typical solar EV charging station implementation is shown in Fig. 1. The major building blocks are self-explanatory.

There is the user side, which basically depicts functionalities visible to the end user. Information exchange and user interaction is taken care of here. It typically consists of a TFT screen with touch-sensing, NFC card readers for authentication or payments, and maybe a Bluetooth interface for more advanced features.

The vehicle is physically connected to any of the output ports—AC slow charge for smaller vehicles and e-rickshaws, AC fast charge for some classes of vehicles and, of course, DC fast charging. The user has to authenticate himself, set charge preferences and wait till the charge session is over. However, more complex functions go on behind the scene, which are controlled and monitored by the central

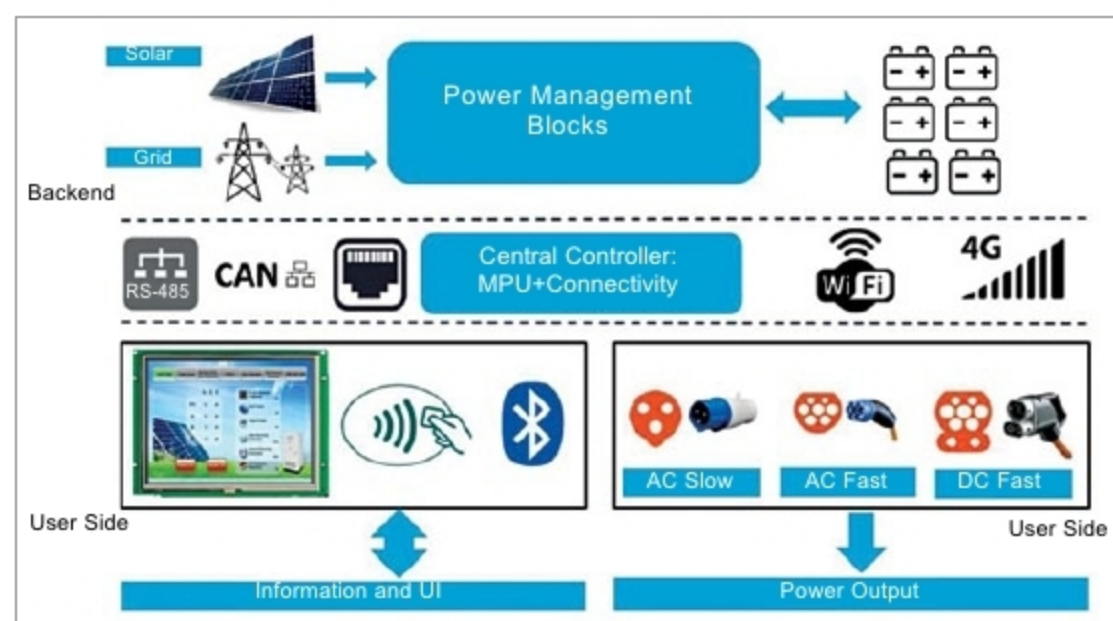


Fig. 1: Solar EV charging station functional blocks